EcoLoop

Circular material systems and zero-cost digital architecture.

Swami Sant Dass Public School, Jalandhar

The Packaging Trap: Multi-Layer Plastics

Why standard municipal recyclers throw clean packaging straight into landfills.

The Material Reality

- **Bonded Metal & Plastic:**
  Chips and snack packets laminate aluminum foil between plastic film. Separating those layers is chemically difficult and commercially unprofitable.
- **100% Landfill Destination:**
  Because municipal plants cannot process multi-layer packaging, almost all snack wrappers are burned or dumped into local landfills.
- **Clean Waste in Schools:**
  Canteens and classrooms generate sacks of clean snack wrappers daily with zero realistic collection logistics.

The EcoLoop Approach

- **Skip Separation, Heat-Compress:**
  Instead of separating the foil, we partner to heat-press whole wrappers into dense boards that replace plywood.
- **Unified Circular Model:**
  Unifying scattered school green crafts and outsourced heavy materials into one clean system.
- **Digital Transparency:**
  Building an online catalog so buyers can see exact specifications and request real samples.

The Physical Loop: In-House Crafts & NGO Partner

What we make directly in school, and what we produce with our partner NGO.

- **Student-Made: Seed Paper**
  Handmade plantable paper embedded with native seeds, crafted in school.
- **Student-Made: Bioenzymes**
  Citrus fruit peels fermented into natural multi-purpose cleaning liquids.
- **Student-Made: Uniform Bags**
  Durable tote bags stitched from discarded school uniform fabrics.
- **Student-Made: Eco-Bricks**
  PET bottles tightly packed with clean plastic wrappers for simple construction.
- **Student-Made: Bottle Planters**
  Upcycled beverage containers for school corridor nursery gardening.
- **NGO Partnership: MLP Sheets**
  Heavy-duty boards heat-pressed from snack wrappers using hydraulic machinery.



Reality Check: How It Started vs How It Is Going



`[Meme: Teachers thinking we'd just pick up litter vs Building an entire brand & platform]`

Clown applying makeup / Roll Safe / SpongeBob presentation

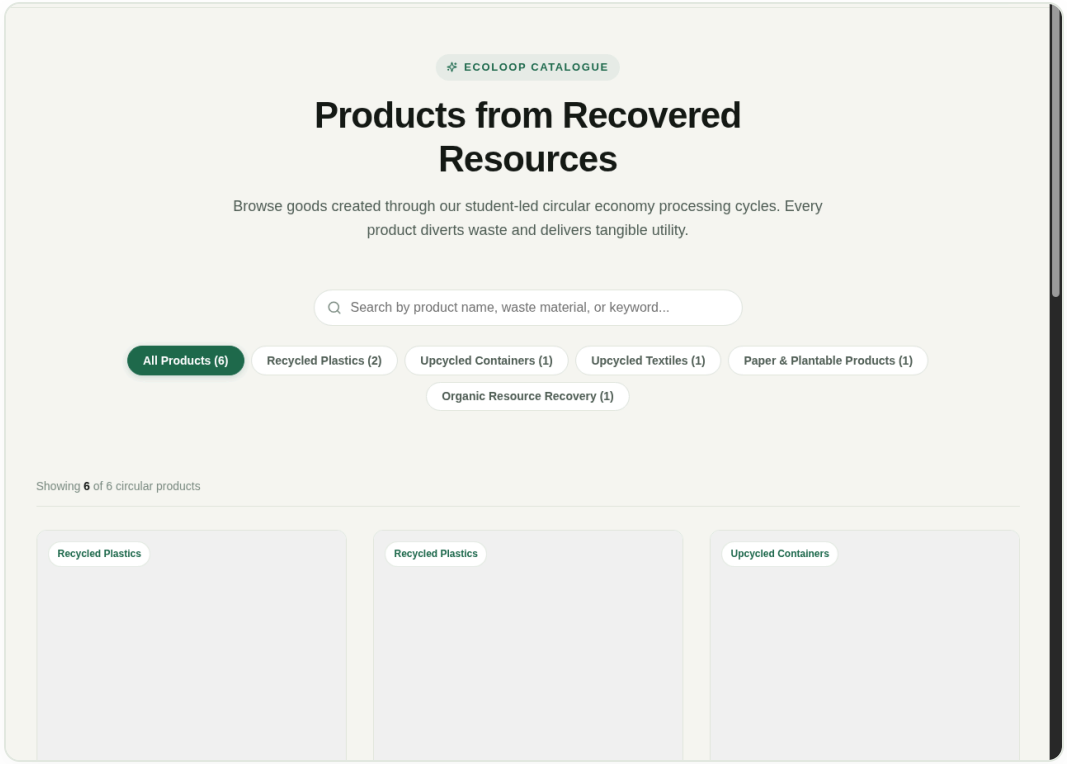
Started as scattered classroom craft projects. Turned into a unified circular brand with external partnerships and custom software.

Why Build a Digital Platform?

Transforming school crafts into verifiable, specification-backed utility products.

Platform Objectives

- **Engineering Credibility:**
  Publishing actual dimensions, material density, thermal ratings, and production lead times so buyers take recycled goods seriously.
- **Direct Inquiry Routing:**
  Giving institutions and partners a friction-free way to query available stock and request physical samples.
- **Connecting School to Outside World:**
  Taking activities happening inside classrooms and presenting them professionally to external organizations.



The Zero-Cost Backend: Built on Google Sheets

Handling live incoming inquiries without paid servers or corporate credit cards.

STEP 01

React Form

User fills in their name, product choice, and notes. The browser validates all inputs locally.

→

STEP 02

Apps Script Webhook

A free Google Apps Script webhook receives the data payload securely in the cloud.

→

STEP 03

Sheets Database

Appends the inquiry row into a live Google Sheet and immediately triggers an email alert.

Monthly Infrastructure Bill

No credit cards, zero server maintenance, and reliable for real order leads.

$0.00 / month

Reality Check: The Cloud Bill Dilemma



`[Meme: Paying $50/month for cloud databases vs High schoolers using a free Google Sheet]`

Buff Doge vs Cheems / Who would win

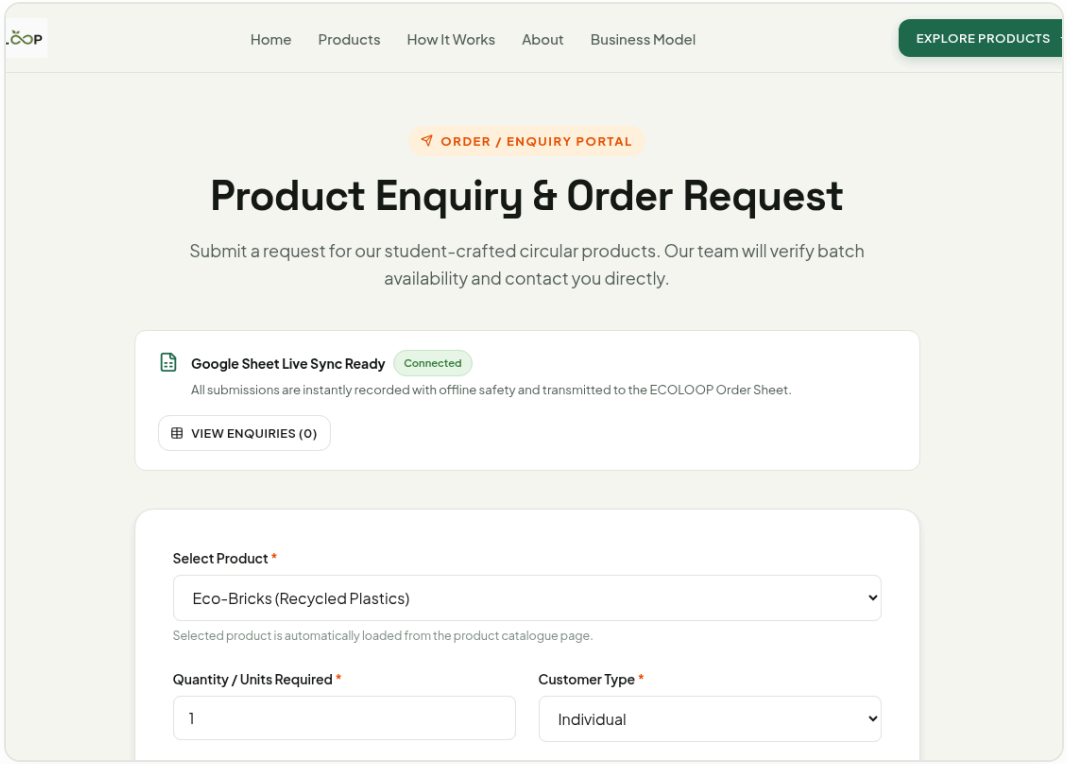
Why spend monthly money on complex database servers when a free Google Sheet captures all your orders?

The Inquiry Engine: An Honest Review

Built in a single night sprint: functional, basic, and with plenty of room to improve.

The Reality of Our MVP

- **Built in One Night:**
  Put together in a single late-night sprint to give the exhibition judges a working demo to test.
- **Basic but Working:**
  Automatically pre-populates product details and logs records straight into our backend without paying for services.
- **Room for Improvement:**
  Needs automated inventory deduction, better error states, and order tracking. It's an MVP, not final software.



How I Built It Without Knowing React

Moving from typing boilerplate code to designing systems with AI agents.

The Traditional Hurdle

- Getting stuck on missing brackets, package version mismatches, and unfamiliar framework setup.
- Spending entire evenings digging through Stack Overflow threads instead of building real features.
- Giving up on software ideas simply because you haven't spent 6 months memorizing syntax.

Developer as Architect

- You specify the data models, visual style, and constraints in plain language.
- An autonomous agent creates components, connects routes, and modifies multiple files.
- Your primary job becomes quality control, testing user flows, and refining the architecture.

Chatbot vs Agent Workspace

The difference between copy-pasting snippets and letting an agent build inside your files.

ChatGPT Web Window

[DROP CHATGPT SCREENSHOT HERE]

- Copy-pasting snippets back and forth
- Forgets project context easily
- Cannot see or edit your local project files

Agent Workspace (Antigravity / OpenCode)

[DROP ANTIGRAVITY SCREENSHOT HERE]

- Reads and edits real repository files
- Understands full folder structure and context
- Runs terminal builds and tests autonomously

How Working with an Agent Actually Looks

A practical 4-step loop of specifying, editing, testing, and self-repairing.

STAGE 01

Write the Spec

Describe what you want in plain language: colors, component layout, and behavior rules.

STAGE 02

Agent Edits Files

The agent inspects your repository, writes components, and updates project files directly.

STAGE 03

You Test Locally

You run the app in your browser, click the buttons, and see what works or what broke.

STAGE 04

Agent Fixes Bugs

You paste the error back to the agent; it reads the log and repairs the broken code.

Replicating This Workflow: OpenCode

An open-source terminal agent that gives you the same autonomous capabilities.

Why OpenCode for Students?

- **100% Free & Open Source:**
  Created by Anomaly (GitHub: `anomalyco/opencode`). Zero subscriptions or vendor lock-in.
- **Lives in Your Terminal:**
  Reads your project files, edits code, and runs local commands just like specialized agent harnesses.
- **Dual Execution Modes:**
  Plan Mode reviews architecture safely; Build Mode executes real code changes.

```
// Quick Installation in 60 Seconds

$ npm install -g @opencode/cli
$ cd ./my-project
$ opencode

Runs on macOS, Linux, and Windows WSL.
```

Free AI Models: Zero Cost for Students

Practical options you can use tonight without paying for subscriptions.

Free Cloud APIs

- **Google Gemini API:**
  Sign in to `aistudio.google.com` for a free key. Generous free limits for fast coding models.
- **Groq Cloud:**
  Ultra-fast free inference on open models like Llama 3.3 via `console.groq.com`.
- **GitHub Models:**
  Free student access to preview models inside VS Code and GitHub Codespaces.

100% Offline with Ollama

- **Private & Local:**
  Download `ollama.com`. Runs entirely on your laptop hardware without needing internet.
- **Recommended Model:**
  `qwen2.5-coder:7b` delivers strong coding reasoning on standard student laptops.
- **Zero Latency:**
  Fast responses directly on your machine with complete data privacy.

Reality Check: AI Confidence vs Reality



`[Meme: AI confidently saying 'all bugs fixed' vs Clicking the button and the console explodes]`

Spider-Man pointing / Buff Doge / Two Buttons

Prompting is only 20% of the work. Testing whether the buttons actually work is the other 80%.

Thank You.

Scan the code to access the complete open-source repository, live demo, and student setup guide.



`github.com/Harman209/ecoloop`

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